

ASSIGNMENT 1

Given Table
EMPLOYEE

EMPNO ENAME SAL DEPTNO DNAME LOC

Current Normal Form

This table is in **First Normal Form (1NF)** because every column contains only one value and there are no repeating groups.

But the table is **not in Third Normal Form (3NF)**.

The reason is that **DNAME** and **LOC** depend on **DEPTNO**, not directly on **EMPNO**. This is known as **transitive dependency**.

Problems in the Table

- Department name is repeated for every employee.
- Department location is also repeated.
- If the department location changes, it has to be updated in many rows.
- This increases duplicate data and may lead to update, insert, and delete anomalies.

Converting to 3NF

EMPLOYEE

EMPNO (PK) ENAME SAL DEPTNO (FK)

DEPARTMENT

DEPTNO (PK) DNAME LOC

Conclusion

By separating the department details into another table, duplicate data is removed and the database becomes easier to maintain. Now the tables satisfy **Third Normal Form (3NF)**.

Assignment 2

Given Table

STUDENT

ROLLNO NAME AGE EXAM MARKS GRADE

Current Normal Form

This table is also in **First Normal Form (1NF)** because all the values are atomic.

However, one student can give many exams, so the same student's details will be repeated.

The primary key is **(ROLLNO, EXAM)**.

Problems in the Table

- **NAME** and **AGE** depend only on **ROLLNO**, not on the complete primary key.
- **GRADE** depends on **MARKS**, which creates transitive dependency.
- Student information gets repeated for every exam.

Converting to 3NF

STUDENT

ROLLNO (PK) NAME AGE

EXAM_RESULT

ROLLNO (FK) EXAM MARKS

Primary Key = (ROLLNO, EXAM)

GRADE

MARKS GRADE

(Grade can also be calculated instead of storing it.)

Conclusion

After dividing the data into different tables, duplicate student information is removed. The table no longer has partial or transitive dependency and is now in **Third Normal Form (3NF)**.

Assignment 3

Given Table

EMPLOYEE_PROJECT

EMPNO PROJECT_NO NO_OF_DAYS CUSTOMERNAME

Primary Key = (EMPNO, PROJECT_NO)

Problem in the Table

The customer name depends only on **PROJECT_NO**, not on both **EMPNO** and **PROJECT_NO**.

This is called **partial dependency**, so the table is **not in Second Normal Form (2NF)**.

Problems

- Customer name is repeated for every employee working on the same project.
- More storage is used because of duplicate data.
- Updating customer details becomes difficult.
- It may also create insert and delete anomalies.

Converting to 2NF

EMPLOYEE_PROJECT

EMPNO (PK) PROJECT_NO (PK) NO_OF_DAYS

PROJECT

PROJECT_NO (PK) CUSTOMERNAME

Conclusion

After moving the customer information into a separate table, duplicate data is removed and the table satisfies **Second Normal Form (2NF)**.